Cost of Living Index Number

$$= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{1000}{758} \times 100 = 131.926$$

Example 9.16

Calculate the cost of living index number for the year 2015 with respect to base year 2010 of the following data.

Commodities	Number of Units (2010)	Price (2010)	Price (2015)
Rice	5	1500	1750
Sugar	3.5	1100	1200
Pulses	3	800	950
Cloth	2	1200	1550
Ghee	0.75	550	700
Rent	12	2500	3000
Fuel	8	750	600
Misc	10	3200	3500

Solution:

Here the base year quantities are given, therefore we can apply Aggregate Expenditure Method.

Commo-dities	Number of Units (2010) q_0	Price (2010) p_0	Price (2015) p_1	p_0q_0	p_1q_0
Rice	5	1500	1750	7500	8750
Sugar	3.5	1100	1200	3850	4200
Pulses	3	800	950	2400	2850
Cloth	2	1200	1550	2400	3100
Ghee	0.75	550	700	412.5	525
Rent	12	2500	3000	30000	36000
Fuel	8	750	600	6000	4800
Misc	10	3200	3500	32000	35000
			Total	84562.5	95225

Table 9.15

Cost of Living Index Number

$$= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{95225}{84562.5} \times 100 = 112.609$$

Hence, the Cost of Living Index Number for a particular class of people for the year 2015 is increased by 12.61 % as compared to the year 2010.

Example 9.17

Calculate the cost of living index number by consumer price index number for the year 2016 with respect to base year 2011 of the following data.

Commo-dities	Price		Quantity
	Base year	Current year	
Rice	32	48	25
Sugar	25	42	10
Oil	54	85	6
Coffee	250	460	1
Tea	175	275	2

Solution:

Here the base year quantities are given, therefore we can apply Aggregate Expenditure Method.

Commo-dities	Price		Quantity (q_0)	p_0q_0	p_1q_0
	Base year (p_0)	Current year (p_1)			
Rice	32	48	25	800	1200
Sugar	25	42	10	250	420
Oil	54	85	6	324	510
Coffee	250	460	1	250	460
Tea	175	275	2	350	550
			Total	1974	3140

Table 9.16

Cost of Living Index Number

$$= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{3140}{1974} \times 100 = 159.0679$$

Hence, the Cost of Living Index Number for a particular class of people for the year 2016 is increased by 59.0679 % as compared to the year 2011.

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